Unsupervised Domain Adaptation LiDAR Semantic Segmentation in Adverse Weather

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Abstract

Collecting fine-grained LiDAR annotations in adverse weather is extremely costly for autonomous driving. Unsupervised Domain Adaptation (UDA) addresses this challenge by training models on clear-weather labeled data and adapting them to unlabeled adverse-weather domains. However, severe domain gaps caused by weather-induced degradation make this adaptation highly challenging. This work reproduces and extends UniMix, the first method directly targeting the LiDAR domain gap between clear and adverse weather. Prior to UniMix, existing approaches relied on semantic-guided mixing or 3D–2D–3D projection pipelines, both of which struggle to handle weather-induced distortions or suffer from substantial information loss. UniMix mitigates these issues by introducing a physically based weather simulation to construct a Bridge Domain and proposing Universal Mixing across spatial, intensity, and semantic dimensions. However, its spatial mixing ignores pseudo-label reliability, allowing low-confidence target regions to introduce noise during training. To address this limitation, we propose a confidence-aware spatial mixing strategy that restricts mixing to high-confidence pseudo-labeled regions, providing more reliable target supervision. Experimental results show that UniMix achieves a +1.8% mIoU improvement over the baseline, while our method achieves a +2.7% improvement.

1. Introduction

LiDAR semantic segmentation aims to predict a semantic class label to each point in a 3D LiDAR point cloud. In autonomous driving, LiDAR semantic segmentation enables more accurate 3D scene understanding by providing fine-grained environmental perception. However, annotating LiDAR data is extremely costly [9], even more so than pixel-wise annotation in RGB images, as every point in the 3D point cloud must be labeled, requiring substantial time and labor.

To address this challenge, Unsupervised Domain Adaptation (UDA) has been widely explored. UDA trains a model on data that is easier and cheaper to annotate, and then adapts it to datasets where annotation is difficult or expensive. However, even when both domains represent road-driving scenarios, significant domain shifts exist, for example, between road data collected in different countries [7] or between clear-weather and adverse-weather conditions [11]. In RGB images, such domain gaps often arise from differences in style [4] or contextual information [1]. In contrast, the domain discrepancy in LiDAR data, especially between clear and adverse weather, is far more severe due to additional factors such as weather-induced light scattering, spatial noise, and changes in reflection intensity. [10, 11] These properties make it substantially more challenging to reduce the domain gap in LiDAR segmentation.

Existing approaches for reducing LiDAR domain gaps generally follow two major directions. (1) Mixing point clouds from the two domains based on semantic information [5], enabling the model to observe both source and target distributions simultaneously. This is an approach inspired by successful mixing strategies in RGB-based UDA [6]. (2) Projecting 3D point clouds into 2D, applying well-established RGB UDA methods, and then reprojecting the results back to 3D [2, 3].

While mixing-based approaches are conceptually effective, they typically rely on semantic information to combine the domains, which limits their ability to handle the most critical weather-induced discrepancies such as spatial noise and intensity degradation. On the other hand, 3D–2D–3D projection methods [3] benefit from mature RGB UDA techniques, but inherently suffer from information loss during projection, necessitating additional corrective mechanisms.

To resolve these limitations, UniMix [11] introduces two key components:

- A Bridge Domain, generated by applying physically realistic weather simulations to the

source domain. The model is first adapted from the clear-weather source domain to the Bridge Domain, and then from the Bridge Domain to real adverse-weather data in a two-stage learning process, effectively mitigating weather-induced domain gaps.

- Universal Mixing, which combines point clouds across spatial, intensity, and semantic dimensions to improve robustness against weather-induced light scattering and intensity variations, and exposing the model to a broader range of target-domain point distributions.

In this project, we focus on reproducing UniMix [11], which is the first representative method addressing the domain gap between clear and adverse weather, and validating its unspecified components through ablation studies. Furthermore, we identify a limitation of UniMix [11] and propose an extended method to address it:

- UniMix [11] performs spatial mixing across the entire point cloud, without considering the confidence of pseudo-labels. We identify this issue as a key limitation of UniMix [11] and propose a confidence-aware spatial mixing strategy that filters out low-confidence pseudo-label regions. This ensures that only reliable target points contribute to mixing, thereby improving adaptation performance under adverse weather conditions.

2. Related Works

Unsupervised Domain Adaptation (UDA) aims to transfer knowledge learned from a labeled source domain to an unlabeled target domain, thereby mitigating the distribution shift between the two domains. In semantic segmentation, UDA is widely used to reduce the substantial domain gap that arises in scenarios such as Synthetic-to-Real adaptation and Clear-to-Adverse Weather adaptation.

Autonomous driving involves complex 3D scenes, making point-level annotation of real-world road data extremely costly [9]. Consequently, models are often trained on synthetic LiDAR data, where annotation is comparatively easier, and then adapted to real environments. However, synthetic LiDAR data does not accurately capture the physical sensing properties of real LiDAR systems, including reflectance-based intensity, noise patterns, and scanning geometry [11]. These discrepancies create a severe geometry–intensity distribution gap, which is often much larger than the style discrepancy commonly observed in RGB image domains [1, 4].

Moreover, the domain shift between clear weather and adverse weather represents another critical challenge in real-world autonomous driving [11]. Under adverse weather conditions, LiDAR point distributions become irregular due to light scattering, backscatter effects, made sparse 3D point clouds, and increased spatial noise [11]. While RGB-based UDA techniques can alleviate weather-related gaps through photometric augmentations or style transformations [8], such transformations cannot be directly applied to LiDAR point clouds. Since weather-induced intensity distributions are highly nonlinear and do not follow simple statistical forms, aligning source and target distributions becomes especially difficult.

Some studies mitigate the domain gap by projecting point clouds into 2D representations and adopting RGB-based UDA methods [10], or by performing domain mixing based on semantic patches. However, these approaches do not adequately capture weather-induced changes such as intensity degradation, spatial noise, or increased sparsity.

UniMix [11] addresses these limitations through a stepwise clear→bridge→adverse domain transition strategy that progressively reduces the extreme weather-induced domain gap. Furthermore, UniMix [11] introduces a universal mixing strategy that integrates Spatial, intensity, and semantic, extending prior semantic mixing approaches to effectively learn weather-robust and domain-invariant representations under adverse conditions.

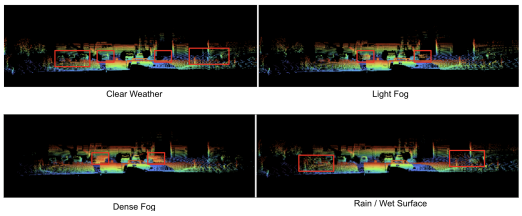

Figure 1. **Bridge Domain Result** Compared to light fog, dense fog exhibits higher scattering and attenuation coefficients, resulting in increased noise and fewer valid returns, which makes the point cloud noticeably more sparse. In fog conditions, backscatter signals tend to concentrate near the LiDAR sensor in the close-range region, whereas in rain, scattering occurs primarily due to individual raindrops, causing the noise to appear more spatially irregular and widely dispersed throughout the scene.

3. Methods

In the UDA, the goal is to adapt a neural network f_θ trained on a labeled source domain $D_S = \{X_S, Y_S\}$ to an unlabeled target domain $D_T = \{X_T\}$. The source domain consists of source point clouds $X_S = \{x_S\}_{i=1}^{N_S}$ with corresponding labels point clouds $Y_S = \{y_S\}_{i=1}^{N_S}$, while the target domain contains only target point clouds $X_T = \{x_T\}_{i=1}^{N_T}$ without annotations. In this work, we reduce the domain gap between the source and target domains by adopting an online self-training strategy, following written in UniMix [11] paper.

In this work, our goal is to implement the methodology described in the UniMix [11] paper as faithfully as possible. For training-related parameters that are not explicitly mentioned in the original paper, such as the number of epochs, learning rate, optimizer, and pseudo-label threshold, we follow the settings used in the baseline method, CoSMix [5]. All other parameters that are not directly discussed in the UniMix [11] method are detailed in our respective subsections.

Furthermore, to address the issue in UniMix's spatial mixing [11] where target regions containing low-confidence pseudo labels may be mixed into the source, we propose a high-confidence pseudo-label–based spatial mixing strategy. We also extend UniMix [11] so that allow controlling the approximate proportion of target points mixed into the source during both spatial mixing and intensity mixing. Our overall pipeline can be seen in Fig. 2.

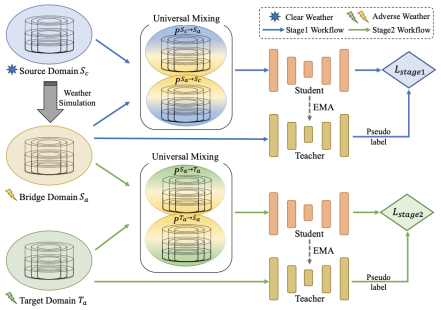

Figure 2. **Overall Pipeline** This follows the same pipeline as UniMix, and the figure is taken from the original UniMix paper.

3.1. Bridge Domain

Bridge Domain reproduces four adverse weather conditions—rain, snow, light fog, and dense fog—on the source domain using physically valid weather simulation. Rain and fog consist of small water droplets, while snow is composed of ice particles with heterogeneous sizes and shapes, causing strong scattering when interacting with laser beams. These airborne scatterers partially scatter and attenuate the laser energy, weakening or distorting the actual reflected signal that LiDAR receives from object surfaces.

LiDAR constructs a point cloud by emitting laser pulses and measuring the time-of-flight (ToF) and intensity of the returned signal. The distance R is computed using the standard time-of-flight model:

$$R = \frac{c\,\Delta t}{2}, \tag{1}$$

where c is the speed of light and Δt denotes the round-trip travel time of the laser pulse. However, under adverse weather, scattered, refracted, or partially reflected light can alter the received waveform, leading to inaccurate ToF estimation and ultimately causing spatial distortions in the reconstructed point cloud.

Furthermore, LiDAR sensors measure the returned intensity from the object surface. Absorption, scattering, and refraction in adverse weather reduce the amount of energy that reaches the sensor, resulting in lower measured intensity values. This degradation in intensity often leads to additional spatial inconsistencies and increased sparsity in the resulting point cloud.

In this project, we generate a Bridge Domain through physically based weather simulation, producing four types of bridge-domain point clouds corresponding to each adverse weather condition. This process can be formulated as: $B_p = \Phi(S_p)$, where S_p denotes the source point cloud and B_p represents the resulting Bridge Domain point cloud.

Since the original UniMix [11] paper does not provide specific parameters for weather-dependent scattering, such as extinction coefficients, we use the default simulation settings for light fog, snow, and rain. For dense fog, the paper also does not specify the required scattering strength. Therefore, we approximate a stronger degradation by increasing the attenuation of the light fog configuration by an additional 15% as a practical heuristic to simulate more severe visibility loss. The generated Bridge Domain results can be seen in Fig. 1.

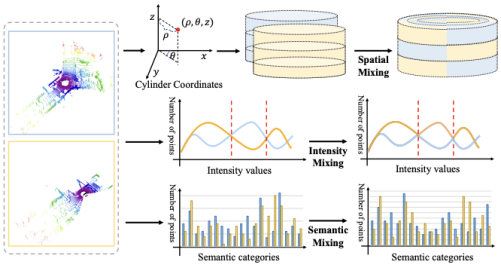

Figure 3. **Universal Mixing Pipeline** This follows the same Universal Mixing pipeline as UniMix, and the figure is taken from the original UniMix paper.

3.2. Universal Mixing

Universal Mixing enhances robustness to the spatial and intensity distribution shifts that occur under adverse weather conditions. In our framework, mixing is not performed directly between the source and target domains; instead, it is carried out between the Source and Bridge Domains, and between the Bridge and Target Domains. For these operations, the Bridge and Target domains use their corresponding pseudo-labels.

Let the domains be defined as follows: $S = (P^S, Y^S)$, $B = (P^B, \hat{Y}^B)$, $T = (P^T, \hat{Y}^T)$, where $\hat{Y}^B$ and $\hat{Y}^T$ denote

the pseudo-labels of the Bridge and Target domains.

Using the Universal Mixing operator Ψ, the mixed intermediate samples are obtained as:

$$\begin{aligned}(P^{S\to B},\, Y^{S\to B}) &= \Psi(S, B),\\ (P^{B\to T},\, Y^{B\to T}) &= \Psi(B, T).\end{aligned} \tag{2}$$

Universal Mixing Pipeline can be seen in Fig. 3.

Spatial Mixing Since LiDAR sensors acquire scenes in a cylindrical manner, performing spatial mixing directly in the 3D Cartesian coordinate space may break the spatial alignment of the scans. Therefore, we first transform the 3D Cartesian coordinates (x, y, z) into cylindrical coordinates (ρ, θ, z) as follows:

$$\rho = \sqrt{x^2 + y^2}, \qquad \theta = \tan^{-1}\left(\frac{y}{x}\right), \tag{3}$$

where ρ is the radial distance from the z-axis and θ denotes the azimuth angle on the sensor's horizontal plane with respect to the x-axis.

Using the transformed coordinates, we partition the space into non-overlapping 3D blocks for spatial mixing. The spatial mixing mask is defined as:

$$M_{\text{Sp}} = \left\{ (\rho, \theta, z) \;\middle|\; \begin{array}{l} (\rho_h < \rho < \rho_h + \Delta\rho) \\ \cap\ (\theta_i < \theta < \theta_i + \Delta\theta) \\ \cap\ (z_j < z < z_j + \Delta z) \end{array} \right\}. \tag{4}$$

where $\Delta\rho$, $\Delta\theta$, and Δz determine the block size along each dimension, and ρ_h, θ_i, and z_j are randomly sampled within the valid ranges. The effect of different interval sizes is analyzed in our ablation study.

Up to this point, our spatial partitioning strategy follows UniMix [11] . However, to introduce our high-confidence pseudo-label–based spatial mixing, we first evaluate the pseudo-label quality before constructing the final spatial mixing mask. Let $p_i \in \mathbb{R}^C$ denote the class probability vector of the pseudo label at point i, where C is the number of semantic classes. We define the confidence of point i as:

$$c_i = \max_c p_{i,c}. \tag{5}$$

Given a confidence threshold τ, we construct a high-confidence mask:

$$\mathcal{H} = \{i \mid c_i \geq \tau\}. \tag{6}$$

We then constrain spatial mixing to only those regions that are both spatially selected and high-confidence:

$$M_{\text{Sp}}^{\text{hc}} = M_{\text{Sp}} \cap \mathcal{H}, \tag{7}$$

and only the points within $M_{\text{Sp}}^{\text{hc}}$ are exchanged between domains during spatial mixing.

Furthermore, we allow for controlling the approximate proportion of target points involved in spatial mixing by adjusting the number of 3D blocks selected from the target (or Bridge) domain. Let K denote the total number of spatial blocks and K_T the number of blocks sampled from the target side. The resulting target mixing ratio can be approximated as:

$$r = \frac{K_T}{K}. \tag{8}$$

We empirically study the effect of varying r in our ablation experiments.

This high confidence pseudo-label–based strategy demonstrates improved robustness compared to the original UniMix [11] approach, and the results of our spatial mixing can be observed in Fig. 4.

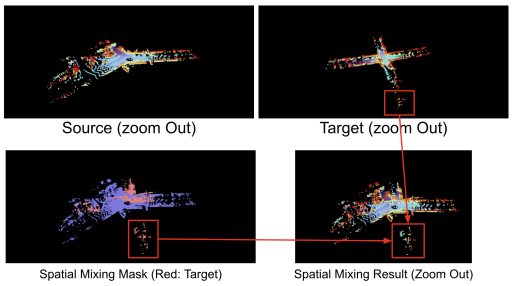

Figure 4. **Spatial Mixing** The source corresponds to the clear-weather domain, while the target is the Bridge Domain. In the spatial mask, the red regions indicate the areas selected from the Bridge Domain. In addition to the regions highlighted by the red bounding box, all red-masked areas are taken from the Bridge Domain, representing the result of selecting three 3D blocks from the Bridge Domain.

Intensity Mixing. For intensity mixing, we follow the original UniMix [11] formulation and additionally introduce a mechanism that allows controlling how much of the target domain is mixed, similar to our extension in spatial mixing. Since the simulated Bridge Domain intensities were scaled by a factor of 255, we first normalize them to the range $[0, 1]$ to match the real source-domain intensity scale.

After normalization, points are exchanged based on their intensity values. Specifically, points whose intensities fall

within a selected interval are swapped between domains: $M_{\text{In}} = \{ I_m < I_i < I_m + \Delta I \}$, where ΔI is a predefined intensity interval and I_m is a randomly sampled intensity value within the valid range. This procedure helps reduce the domain gap caused by intensity distribution shifts.

Semantic Mixing. Semantic mixing also follows the original UniMix [11] formulation, which is fully consistent with the baseline method CoSMix [5]. Using high-confidence pseudo-labels, we ensure that classes not selected from the source domain are instead selected from the target domain. Formally, let $\hat{Y}^T$ denote the pseudo-labels of the target domain and C_S denote the set of semantic classes present in the selected source region. The semantic mixing mask is defined as: $M_{\text{Se}} = \{ p \mid \hat{Y}^T(p) \notin C_S \}$, where points whose pseudo-labels fall outside the source class set C_S are taken from the target domain. This encourages balanced class-level mixing and improves robustness under class distribution shifts.

Overall Universal Mixing To mitigate the domain gaps arising from spatial, intensity, and semantic discrepancies, we apply all three mixing strategies by defining $\mathbf{M}_S \in \{M_S^\sigma \mid \sigma \in \{\text{Sp}, \text{In}, \text{Se}\}\}$ and $\mathbf{M}_T \in \{M_T^\sigma \mid \sigma \in \{\text{Sp}, \text{In}, \text{Se}\}\}$. This follows the original UniMix formulation.

4. Experiments

4.1. Implementation Details

Following UniMix [11] , we evaluate our UDA performance on two benchmark settings: SemanticKITTI → SemanticSTF and SynLiDAR → SemanticSTF. Here, SemanticKITTI represents a clear-weather real-world LiDAR dataset, SynLiDAR is a synthetic source dataset, and SemanticSTF serves as the adverse-weather target domain.

To enable training on two RTX 4070 GPUs, we additionally implement Distributed Data Parallel (DDP). The source-only results are obtained on a single RTX 2080 Ti with a batch size of 1, whereas all UDA experiments are conducted with a batch size of 3.

To improve generalization, we adopt the SGD optimizer with a learning rate of 0.01 and apply a warmup schedule. Warmup strategies have been shown to enhance generalization in RGB-based UDA settings, and we observe similar benefits in LiDAR UDA experiments.

5. Results

5.1. UDA in Real Road Data

To evaluate the UDA performance under clear-to-adverse weather conditions in the real road dataset, we conducted experiments on the SemanticKITTI→SemanticSTF benchmark. Since CosMix [5] serves as our baseline, we compare CosMix [5], UniMix [11] , and our proposed high-confidence pseudo-label–based mixing strategy in Table 1.

Both CosMix [5] and UniMix [11] were evaluated using the same Source-only model with an mIoU of 24.4%. CosMix [5] improves the performance to 28.4% (+4.0% over Source-only), while UniMix [11] achieves 37.2% (+12.8% over Source-only). The substantial gain from UniMix [11] is expected, as it performs universal mixing across semantic, spatial, and intensity dimensions and additionally leverages the Bridge Domain.

In our experiments, the Source-only model achieves 22.9% mIoU, and our final method reaches 38.1%, resulting in a +15.2% improvement. As shown in the ablation study, using a smaller batch size of 2 yields 36.9% (+14.0%), indicating that the performance gain cannot be attributed solely to batch size. Instead, these results confirm that our proposed high-confidence pseudo-label–based spatial mixing significantly contributes to the performance improvement.

5.2. UDA in Clear Weather Synthetic Data to Adverse Weather Real Data

To further evaluate the performance gap between synthetic clear-weather data and real adverse-weather conditions, we conduct experiments on the SynLiDAR→SemanticSTF benchmark. This setting is significantly more challenging, as it involves both a synthetic-to-real domain shift and a clear-to-adverse weather shift. As shown in Table 1, the overall performance is noticeably lower than in the real clear-weather→real adverse-weather setting (SemanticKITTI→SemanticSTF).

CosMix [5] and UniMix [11] both use a Source-only model with an mIoU of 15.0%, achieving 22.1% (+7.1%) and 26.7% (+11.7%), respectively. In our experiments, the Source-only model obtains 13.3% mIoU, and our final UDA performance reaches 24.8%, yielding an improvement of +11.5%. This result is obtained using a batch size of 3. Under the assumption that UniMix [11] is evaluated with a batch size of 2, as used in its baseline configuration, UniMix [11] achieves a higher relative gain in this particular setting.

This phenomenon can be attributed to the reduced quality of pseudo labels in the synthetic-to-real adverse-weather scenario. Since spatial mixing introduces a relatively large proportion of target points, low-quality pseudo labels can negatively affect training. We empirically observe that after a certain number of iterations, the mIoU begins to decrease, suggesting that the model gradually collapses by following incorrect pseudo-label guidance—a well-known issue in UDA when pseudo-label quality deteriorates.

However, this also highlights an important aspect of UDA: exposing the model to more target-domain

Method	car	bi.cle	mt.cle	truck	oth-v.	pers.	bi.clst	mt.clst	road	parki.	sidew.	oth-g.	build.	fence	veget.	trunk	terra.	pole	traf.	mIoU
SemanticKITTI→SemanticSTF																				
Source-only	55.9	0.0	0.2	1.9	10.9	10.3	6.0	0.0	61.2	10.9	32.0	0.0	67.9	41.6	49.8	27.9	40.8	29.6	17.5	24.4
CoSMix [5]	65.0	**1.7**	22.1	**25.2**	7.7	33.2	0.0	0.0	64.7	11.5	31.1	0.9	62.5	37.8	44.6	30.5	**41.1**	30.9	28.6	28.4
UniMix [11]	75.3	0.9	44.9	11.7	**13.6**	**38.2**	**50.3**	**31.9**	71.1	15.0	**46.4**	6.5	74.3	**51.0**	49.8	**36.8**	34.4	25.5	28.9	37.2
Our Result (src-only: 22.9)	**77.1**	0.6	**45.2**	14.4	11.8	35.7	47.9	31.8	**73.2**	**17.8**	45.5	**10.3**	**89.7**	38.9	**51.3**	35.7	31.4	**34.0**	**31.6**	**38.1**
SynLiDAR→SemanticSTF																				
Source-only	27.1	**3.0**	0.6	15.8	0.1	25.2	1.8	5.6	23.9	0.3	14.6	0.6	36.3	19.9	37.9	17.9	**41.8**	9.5	2.3	15.0
CoSMix [5]	51.5	0.2	5.0	28.1	0.0	26.5	**17.0**	9.9	20.2	3.6	24.6	2.2	52.6	20.6	47.5	24.3	34.6	28.2	24.1	22.1
UniMix [11]	**73.6**	0.0	**7.9**	26.9	**2.9**	29.1	13.7	**21.8**	38.0	**8.0**	**26.3**	3.4	**56.0**	**21.2**	56.1	**29.6**	38.0	28.2	**26.5**	**26.7**
Our Result (src-only: 13.3)	50.0	0.0	2.4	**29.3**	0.3	**31.8**	15.0	19.6	**41.2**	5.5	21.9	**4.2**	54.7	20.9	**56.3**	25.1	36.3	**31.7**	25.0	24.8

Table 1. Comparison of SOTA domain adaptation methods on SemanticKITTI→SemanticSTF and SynLiDAR→SemanticSTF. SemanticKITTI and SynLiDAR serve as the source domain while SemanticSTF with all four weather conditions serves as the target domain.

Ablation Setting	Config	mIoU
Batch Size	1	35.4
	2	36.9
	3	**38.1**
Spatial Mixing Ratio	10%	36.5
	15%	**38.1**
	20%	36.7
	Paper-like	35.8
Mixing Strategy	Spatial only	35.8
	Intensity only	33.8
	Semantic only	35.2
	All (Paper)	37.2
	All (Ours)	**38.1**
Bridge Domain Weather	Dense Fog	33.8
	Light Fog	35.5
	Rain	35.1
	Snow	**35.8**

Table 2. Ablation study on SemanticKITTI→SemanticSTF. We analyze the impact of batch size, spatial mixing target ratio, mixing strategy, and single-weather bridge domain selection.

structure can improve adaptation performance, but only when pseudo-label quality is sufficiently reliable. Our method combines high-confidence pseudo-label–based spatial mixing with controlled target exposure through multiple 3D blocks. As demonstrated in the real clear-weather→adverse-weather setting, this combination leads to substantial improvements. These results reinforce a typical UDA principle: strong performance arises from the synergy between high-quality pseudo labels and increased target-domain exposure.

5.3. Ablation Study

Our ablation study is presented in Table 2. Under our mixed-domain training setting with noisy pseudo-label supervision, we observe that a larger batch size leads to more stable optimization and improved generalization performance. In particular, a batch size of 3 achieves the best results among all configurations. Regarding the spatial mixing ratio, the best performance is obtained at 15%. Since our framework combines spatial, intensity, and semantic mixing strategies, using an excessively large spatial mixing ratio exposes the model to an overwhelming amount of target-domain data. As the target labels are pseudo-labels rather than ground truth, moderate exposure to the target domain helps improve performance, whereas excessive exposure can cause catastrophic forgetting of the source domain or even lead to model collapse. This observation is further supported by the spatial-mixing-only results, where a higher ratio of 20% achieves the best performance of 35.8%, indicating that an appropriate level of target-domain exposure is crucial for effective adaptation. Finally, we evaluate the effect of the bridge domain by using only one weather condition at a time instead of all four conditions. The snow condition yields the best performance, which is consistent with the observations reported in UniMix [11] . In contrast, fog and rain introduce stronger domain gaps due to atmospheric scattering effects, leading to degraded performance.

6. Conclusion

In conclusion, we demonstrate that mixing across domains, particularly the Universal Mixing strategy proposed in UniMix [11] , is highly effective in reducing the domain gap in LiDAR-based UDA. Furthermore, our results show that incorporating pseudo-label confidence into the mixing process leads to more meaningful and effective mixing. We observe that model performance consistently improves as the model is exposed to an appropriate amount of target-domain samples.

As future work, it would be good to extend this approach by introducing lightweight heuristics to better filter reliable pseudo-label regions. Specifically, combining entropy-based measures with pseudo-label confidence, or leveraging differences in softmax probability distributions(Max Prob1 - Max Prob2), may allow more informative pseudo-label regions to be selected compared to relying on pseudo-label confidence alone.

One limitation of the current study is that, when constructing the dense fog bridge domain, we apply approximately 15% stronger scattering than the light fog setting. A more comprehensive ablation study over varying fog densities would provide deeper insights into how the severity of fog influences the domain gap, which we leave for future investigation.

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